# Albert Ankomah Dodoo

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## SUMMARY

I am a results-driven Machine Learning researcher with expertise in pattern recognition, generative AI, swarm intelligence, and optimization algorithms. I am skilled in using machine learning models such as Vision Transformers (ViT), Generative Adversarial Networks, Graph Neural Networks (GNNs), and Long Short-Term Memory (LSTM) for application in healthcare, cybersecurity, ISRU, and robotics. I am proficient in using research findings from in-depth analysis to support decision-making. I have a proven track record in mentoring undergraduate students interested in machine learning research and its application.

## RESEARCH INTEREST

- Pattern Recognition and Image Analysis
- Trustworthy and Adaptive Multimodal Learning in Dynamic Environments
- Optimisation of Swarm Intelligence and Multi-Agent Systems

## EDUCATION

**University of Ghana** Accra, Ghana
MSc. Computer Science Sep 2024 – Nov 2025
Thesis: Adaptive Memory-Based Cuckoo Search for High-Dimensional Feature Optimization in Non-Stationary Fraud Detection Systems.
Supervisor: Professor Justice Kwame Appati

**University of Ghana** Accra, Ghana
BSc. Computer Science Aug 2019 – Sep 2023
Thesis: Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities.
Supervisor: Professor Justice Kwame Appati

## RESEARCH EXPERIENCE

**Data Intelligence and Swarm Analytics Laboratory (DISAL)** Accra, Ghana
Research Assistant Aug 2022 - Present

- Collaborated with two undergraduate students on conducting research on Vision Transformer-based COVID-19 detection using chest CT scans.
- Researched on custom-based mobile-network model for early blight disease detection in tomatoes.
- Achieved 98% accuracy in malware detection by employing an LSTM model on correlation-based feature descriptors.
- Researched an adaptive memory-based Cuckoo Search algorithm for feature optimization in non-stationary fraud detection scenarios, addressing concept drift.
- Guided six undergraduate students in machine learning research through weekly meetings and collaborative projects.
- Attended and presented research findings at a conference organized by ISSER, focusing on custom-based mobile-network models for tomato disease detection.

**Knoxxi Ghana Limited** Accra, Ghana
Data Scientist (Contract) Oct 2025 – Jan 2026

- Develop and maintain dynamic Apache Superset dashboards tracking KPIs across operations, product, finance, and customer performance to support decision making.
- Partner with business executives and product managers to translate analytics into actionable business insights.
- Deliver bi-weekly presentations of analytical insights and visualizations to key stakeholders, enabling data-driven understanding of business trends and performance.

**Space Copy Inc.** Delaware, United States
STEM Research Intern Jun 2025 – Sep 2025

- Reviewed 50+ peer-reviewed papers on on-device AI, runtime optimization, and explainability for autonomous edge systems.
- Assessed embedded vision and sensor fusion methods in lunar conditions, focusing on optimization trade-offs between accuracy and efficiency.
- Supported grant writing for the NASA CAN on SLM Additive Manufacturing Dual Technologies (MSFC).
- Produced actionable insights for deploying trustworthy AI in space robotics and other mission-critical edge applications.

**University of Ghana Department of Computer Science** Accra, Ghana
Research Assistant Sep 2023 – Oct 2024

- Conducted a systematic literature review on network security and privacy using the PRISMA framework to explore trends since 2020.
- Analysed Vision Transformer (ViT) models for mobile malware detection on Android platforms, improving early detection rates.
- Contributed to a Google-funded HCI Lab project by curating Akan language data to build NLP models for profanity detection in low-resource content moderation.

**Negotiated Benefits Trust Company Limited** Accra, Ghana
Research Intern Oct 2021 – Dec 2021

- Analysed client's data using statistical models and visualization techniques for improved business insights.
- Developed a customer churn prediction model using machine learning, forecasting customer retention risks.

## TEACHING EXPERIENCE

**University of Ghana Department of Computer Science** Accra, Ghana
Teaching Assistant Sep 2023 – Oct 2024

- Led tutorial sessions for 70 students on subjects like Machine Learning, Numerical Methods, and Statistical Models for Data Science to improve their understanding.
- Designed quizzes for 70 students and provided feedback to enhance students' understanding.
- Demonstrated practical applications of statistical concepts such as standardization, z-score, and measures of central tendency through Python code implementations during lectures.
- Organized mentorship sessions, inviting industry professionals to offer career insights and guidance.

## LEADERSHIP & MENTORING

- Research Mentor, Data Intelligence and Swarm Analytics Laboratory (DISAL), 2022 - Present
- Secretary, Computer Science Students' Association (COMPSSA), University of Ghana, 2022 – 2023.

## PUBLICATIONS AND REPORTS

- Appati, K.J., Ziamah, B., Akrofi, H.A., & **Dodoo, A.A.** SARS Detection in Chest CT-Scan Images Using Bootstrapped ViT-B16 Model. Springer, Iran Journal of Computer Science (2025). doi.org/10.1007/s42044-025-00231-1
- **Dodoo, A.A.,** Koranteng, M., Amissah, D.K., & Appati, K.J. Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4219089/v1
- Quaicoo, R., Acheampong, R., Gyamenah, P., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. Adapting Triple BigGAN for Image Detection Tasks: Challenges and Opportunities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4262097/v1
- Tetteh, L.M., Sarfo, O.Y., Amissah, D.K., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. (2023). Enhancing Unsupervised Signature Verification Through Advanced Feature Extraction Techniques. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4249016/v1

## AWARDS, RECOGNITIONS, AND CERTIFICATIONS

- Overall Best Graduating Graduate Student at the Department of Computer Science (December 2025).
- DataCamp Donates X DataGlobal Hub Scholarship (October 2025)
- Overall Best Research Paper Award at the University of Ghana Postgraduate Students' Research Workshop (July 2025)
- Signal Processing (Python) for Neuroscience Practical Course, Udemy (January 2025).
- Large Language Models (LLM) Course, Vilnius University, Lithuania (December 2024).
- Research Fellow (Beginner), Data Intelligence and Swarm Analytics Laboratory (August 2022).

## TECHNICAL SKILLS

Programming Languages: Python | SAS | R | MATLAB | SQL
Machine Learning Models: CNN | LSTM | GRU | GNN | Autoencoders | Vision Transformers | GANs | Large Language Models
Libraries: TensorFlow | Keras | Scikit-learn | Pandas |NumPy | Pytorch | Hugging Face Transformers | NLTK | OpenCV |Seaborn
Visualization Tools: Apache Superset | Power BI
Productivity Tools: Git |Microsoft Office Suites | LaTeX